

INIS: A Family of $\Delta\Sigma$ Modulators With Inherent Spur Immunity When Interacting With a Static Nonlinearity

Alessandro Ravera^{ID}, *Member, IEEE*, Valerio Mazzaro, Marco Storace^{ID}, *Senior Member, IEEE*,
and Michael Peter Kennedy^{ID}, *Fellow, IEEE*

Abstract—Digital $\Delta\Sigma$ modulators (DDSM) are used in applications that require a reduction of the wordlength of a digital signal. In the presence of non-idealities, the quantization error of the DDSM can interact with nonlinearities further along the signal chain and generate spurious tones in addition to excess noise. This paper introduces the INIS family of DDSMs that are inherently immune from nonlinearity-induced spurs. Some representative members of the family are analyzed through simulations; one is demonstrated with a hardware implementation. Their performances are compared with those of other well-known DDSM architectures.

Index Terms—Digital delta-sigma modulator (DDSM), spurious tones, noise folding, nonlinearity.

I. INTRODUCTION

DIGITAL $\Delta\Sigma$ modulators (DDSM) are commonly used in modern systems such as frequency synthesizers and audio/video processing blocks for implementing the operation of *requantization*.

The operation of requantization allows one to reduce the number of quantization levels of a digital signal without a significant loss of information. In the process, a quantization error is generated and effectively added to the output signal. If the requantization is not properly implemented, the resulting quantization error can be detrimental to the noise performance of the system. For this reason, families of conventional DDSMs are designed such that the quantization error is appropriately randomized and spectrally shaped. The former ensures that the power spectral density (PSD) of the quantization error is free from idle tones. This is normally

obtained by using techniques that increase the cycle length of the modulator [1], [2].

Secondly, the PSD of the quantization error is typically high-pass shaped so that the noise is pushed to frequencies outside the signal band of interest. While the DDSM pushes the quantization noise to high frequencies, the quantization error signal can interact with subsequent nonlinearities in the signal chain *after* the DDSM and “fold” this noise back into the signal band. In addition, interaction between the DDSM and the nonlinearity can produce unwanted spurious tones (spurs) in the signal band [3], [4], [5].

In digital-to-analog converters (DACs), the quantization error signal introduced by the DDSM interacts with nonlinearities arising from systematic and random mismatches among circuit elements that are nominally identical. Some works have addressed the problem of noise resulting from DAC distortion by spectrally shaping the error using techniques like mismatch shaping and dynamic element matching [6], [7], [8]. These techniques are typically applied to the output of the DDSM and require the output to be thermometer coded. Consequently, as the resolution of the DAC increases, the hardware needed for these techniques grows exponentially [9], [10].

A class of quantizers with optimal immunity to spurious tones, referred to as the Successive Requantizer (SR), has been introduced in the literature [11], [12], [13]. While DDSMs are typically designed to minimize quantization error, SR relaxes this bound in order to ensure immunity to spurious tones. It follows that the order of the high-pass noise shaping of the quantization error is limited.

In this work, we focus on the periodic component of the noise that is generated by the interaction of the DDSM's output with a static nonlinearity, as in DACs, but we approach the problem in a different manner [14]. The question we address is: can we have a family of DDSMs that are inherently (i.e., without the application of post-processing techniques) immune from nonlinearity-induced spurs and that could produce a smaller quantization error compared to the SR? We provide a positive answer to this question and present a novel family of DDSMs (with the acronym INIS) that are immune from nonlinearity-induced spurs. An INIS-DDSM inherently provides immunity from nonlinearity-induced spurs independently of the static nonlinearity when its output is applied *directly* to that nonlinearity. Additionally, it is shown that INIS-DDSMs outperform the SR in terms of noise power.

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Alessandro Ravera and Marco Storace are with the Department of Electrical, Electronic, Telecommunications Engineering and Naval Architecture, University of Genoa, 16145 Genoa, Italy (e-mail: alessandro.ravera@edu.unige.it; marco.storace@unige.it).

Valerio Mazzaro is with the School of Electrical and Electronic Engineering and the Microelectronic Circuits Centre Ireland, University College Dublin, Dublin 4, D04 V1W8 Ireland, and also with STMicroelectronics, 27100 Pavia, Italy (e-mail: valerio.mazzaro@st.com).

Michael Peter Kennedy is with the School of Electrical and Electronic Engineering and the Microelectronic Circuits Centre Ireland, University College Dublin, Dublin 4, D04 V1W8 Ireland (e-mail: peter.kennedy@ucd.ie).

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Using a brute-force approach (extensive numerical simulations), sufficient conditions for achieving spur immunity [15] are shown to hold. The spur immunity property is further validated through other simulations and experimental results. The analytical proof of this property remains an open issue, a scientific challenge that needs to be addressed in future studies.

The analysis carried out in this paper focuses on DC inputs only. There are two main reasons for this choice. First, in DACs, a DDSM is often employed to requantize a constant or slowly varying digital sequence. Second, if a modulator remains free of quantization tones with DC inputs, it is likely to be tone-free for other input types as well [16].

Reference [17] is a study on the generation of spurious tones when a DDSM is followed by a static nonlinearity. A family of DDSMs with the property of spur immunity when used inside a fractional-N frequency synthesizer is presented in [18]. In that work, the focus is on preventing the generation of spurious tones when the *accumulated* quantization error interacts with a static nonlinearity. In the case of application to DACs, what generates the spurious tones is the interaction between the *non-accumulated* quantization error and a static nonlinearity. Thus, the family introduced in [18] cannot be used. In this paper, we propose a family of DDSMs with properties different from those in [18]; this family has the property of intrinsic spur immunity to static nonlinearities when used within a DAC.

The paper is organized as follows. Section II introduces the INIS-DDSM architecture and discusses its properties. A representative example of an INIS-DDSM driving a nonlinear DAC is presented in Section III and compared with other architectures. In Section IV, an INIS-DDSM experimental demonstrator is implemented using a commercial FPGA and DAC. Section V provides further examples of INIS-DDSMs and further numerical analyses with comparisons. Finally, we draw our conclusions in Section VI.

II. DDSMs WITH IMMUNITY FROM NONLINEARITY-INDUCED SPURS

A. Background

A DDSM with an input $x[n]$ introduces a quantization error, denoted by $e_q[n]$, such that the output $y[n]$ can be expressed as:

$$y[n] = x[n] + e_q[n]. \quad (1)$$

We assume that the input is normalized:

$$x[n] \in [0, 1). \quad (2)$$

The statistical and spectral properties of the quantization error depend on the type and order of the modulator. Without loss of generality, we can consider the model of a multibit error feedback modulator (EFM) shown in Fig 1(a), whose linearized version is shown in Fig. 1(b).

The digital quantizer implements a truncation while generating the integer output $y[n]$. This can be modelled as the introduction of an error signal $e[n]$, which is in general different from $e_q[n]$. The quantizer typically implements a

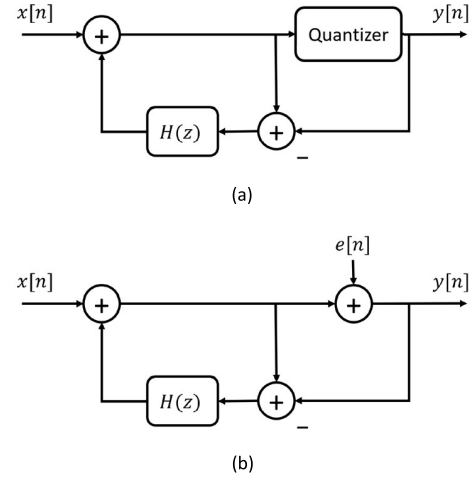


Fig. 1. (a) Block diagram of the EFM. (b) Linearized model.

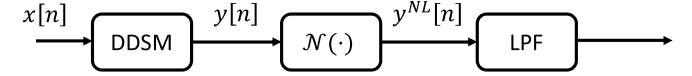


Fig. 2. Signal path comprising a DDSM, a static nonlinearity, and a lowpass filter and related spectra.

floor function, rounding its input to the closest smaller integer value. Therefore, the signal $e[n]$ is such that:

$$e[n] \in (-1, 0]. \quad (3)$$

The Z-transform of the output $y[n]$ is given by:

$$Y(z) = X(z) + NTF(z)\mathcal{E}(z), \quad (4)$$

where $\mathcal{E}(z)$ is the Z-transform of $e[n]$ and $NTF(z)$ is the noise transfer function. Comparing Eqs. (4) and (1), one can notice that:

$$\mathcal{E}_q(z) = NTF(z)\mathcal{E}(z), \quad (5)$$

where $\mathcal{E}_q(z)$ is the Z-transform of the quantization error $e_q[n]$. According to the structure shown in Fig. 1, the transfer function $H(z)$ has the form [19]:

$$H(z) = 1 - NTF(z). \quad (6)$$

The noise transfer function must typically be that of a high-pass filter in order to move the quantization error out of the signal band. Consider a signal path comprising a cascade of a DDSM, a static nonlinearity, and a lowpass filter, as shown in Fig. 2. The signal $x[n]$ is requantized by a DDSM, which introduces the quantization error $e_q[n]$. Then, the signal $y[n]$ is distorted by a nonlinear function and delivered to a continuous-time lowpass filter. Distortion due to the nonlinearity $\mathcal{N}$ causes the signal $y^{NL}[n]$ to exhibit excess folded noise and spurs in the signal band; these are not removed by the filter.

Let the distorted signal be

$$y^{NL}[n] = \mathcal{N}(y[n]), \quad (7)$$

$\mathcal{N}(\cdot)$ where denotes the discrete nonlinear function.

As anticipated in the Introduction, the question we address in this work is the following: can we choose a transfer function $H(z)$ so that the DDSM is guaranteed not to produce

spurs when it interacts with a static nonlinearity? This would avoid the need of applying techniques of spectral shaping to the DDSM output, with a consequent reduction of hardware resources, mainly for high-resolution DACs [9], [10]. In the following, we call such a DDSM an INIS-DDSM.

B. Key Parameters of an INIS-DDSM

In the EFM structure shown in Fig. 1,

$$H(z) = -\sum_{i=1}^l d_i z^{-i}. \quad (8)$$

Therefore, from Eq. (6), the $NTF(z)$ of an l -th order INIS-DDSM has the form:

$$NTF(z) = 1 + \sum_{i=1}^l d_i z^{-i}. \quad (9)$$

The coefficients d_i ($i = 1, \dots, l$) are usually chosen as integers.

Substituting Eq. (9) into Eq. (5), the Z-transform of the quantization error of an INIS-DDSM is given by:

$$\mathcal{E}_q(z) = \mathcal{E}(z) \left(1 + \sum_{i=1}^l d_i z^{-i} \right). \quad (10)$$

The quantization error is expressed in the time domain as:

$$e_q[n] = e[n] + \sum_{i=1}^l d_i e[n-i]. \quad (11)$$

Therefore, the output of the modulator is:

$$y[n] = x[n] + e[n] + \sum_{i=1}^l d_i e[n-i]. \quad (12)$$

The spread of the DDSM output signal $y[n]$ defines the extent of the exposure of $y[n]$ to the nonlinear function. Since $e[n]$ represents the error signal, which results from implementing a floor function (denoted as $\lfloor \cdot \rfloor$), Eq. (12) can be recast as follows

$$\begin{aligned} y[n] &= \left\lfloor x[n] + \sum_{i=1}^l d_i e[n-i] \right\rfloor \\ &= \left\lfloor x[n] + \sum_{d_i > 0} d_i e[n-i] + \sum_{d_i < 0} d_i e[n-i] \right\rfloor. \end{aligned} \quad (13)$$

Given the ranges of $x[n]$ and $e[n]$, defined in Eqs. (2) and (3), and since the coefficients d_i are integers, one obtains that

$$-\sum_{d_i > 0} d_i \leq y[n] \leq -\sum_{d_i < 0} d_i. \quad (14)$$

Therefore, $y[n]$ can assume values out of a set of $(w+1)$ integers, where

$$w = \sum_{d_i > 0} d_i - \sum_{d_i < 0} d_i = \sum_{i=1}^l |d_i|. \quad (15)$$

The higher is w , the greater the level of nonlinearity to which $y[n]$ is potentially exposed.

The quantization error $e_q[n]$ results from $e[n]$ passing through a finite impulse response (FIR) filter with transfer function $NTF(z)$, as is clear from Eqs. (9)–(11). To suppress the low-frequency component of the modulation-related noise, the FIR filter has to be a high-pass filter. As a result, $NTF(z)$ is null for $\omega = 0$, requiring that it has at least one zero at $z = 1$. In other words, the coefficients d_i in Eq. (9) need to satisfy:

$$1 + \sum_{i=1}^l d_i = 0. \quad (16)$$

In particular, the initial order h of high-pass shaping at low frequencies is given by the number of solutions of $\mathcal{E}_q(z) = 0$ at $z = 1$. The higher is h , the lower is the resulting quantization error at low frequencies.

The DAC's nonlinear transfer characteristic applied to $y[n]$ is a discrete function that can assume $(w+1)$ real values. Therefore, a generic nonlinearity $\mathcal{N}(\cdot)$, limited to the range of $y[n]$, can always be expressed as a polynomial function with order w . In other words, for any nonlinearity, we can express $y^{NL}[n]$ as a polynomial function of $y[n]$ with real coefficients α_k , with $k = 1, \dots, w$:

$$\begin{aligned} y^{NL}[n] &= \mathcal{N}(y[n]) \\ &= \alpha_1 (x[n] + e_q[n]) + \alpha_2 (x[n] + e_q[n])^2 \\ &\quad + \dots + \alpha_w (x[n] + e_q[n])^w \end{aligned} \quad (17)$$

As we consider a constant input $x[n]$, $y^{NL}[n]$ can be recast as

$$y^{NL}[n] = \beta_0 + \sum_{k=1}^w \beta_k e_q^k[n], \quad (18)$$

with real coefficients β_k , with $k = 0, \dots, w$.

A sequence $y^{NL}[n]$ is said to be immune to spurious tones up to order u if, for $p = 1, 2, \dots, u$, $(y^{NL}[n])^p$ are free of spurious tones [13].

The Wiener-Khinchin Theorem shows that, if $y^{NL}[n]$ is a wide-sense-stationary random sequence (and we will show later that this is true, see Theorem 2), its PSD is given by the Fourier transform of the autocorrelation $R_{y^{NL}}[k]$ [20]. Based on Eq. (18), we obtain

$$\begin{aligned} R_{y^{NL}}[k] &= \beta_0^2 \\ &\quad + \beta_0 \sum_{p=1}^w \beta_p E[e_q^p[n-k]] + \beta_0 \sum_{p=1}^w \beta_p E[e_q^p[n]] \\ &\quad + \sum_{p_1=1}^w \sum_{p_2=1}^w \beta_{p_1} \beta_{p_2} E[e_q^{p_1}[n] e_q^{p_2}[n-k]]. \end{aligned} \quad (19)$$

This means that for the spur immunity, the following properties hold.

Property 1: For a constant input $x[n]$, the output $y^{NL}[n]$ is said to be immune to spurious tones up to order u , if the sequences $e_q^p[n]$, for $p = 1, 2, \dots, u$, are immune to spurious tones and the sequences $e_q^{p_1}[n]$ are uncorrelated with $e_q^{p_2}[n]$, where p_1 and p_2 are distinct integers in $1, 2, \dots, u$.

Property 2: In order to achieve spur immunity for any possible nonlinearity, an INIS-DDSM needs to provide

$$u \geq w. \quad (20)$$

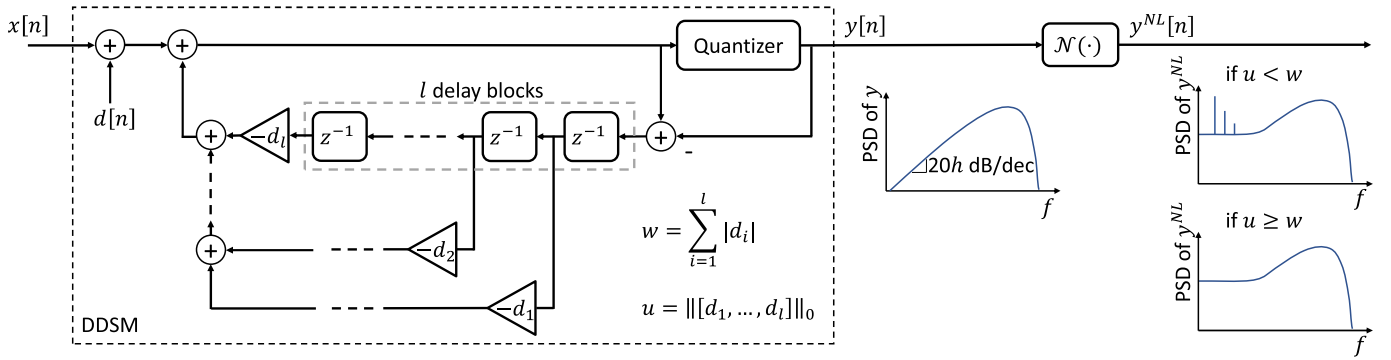


Fig. 3. Signal path of the DAC, comprising a DDSM and a static nonlinearity, and key parameters (l , h , w , u).

In summary, the key parameters that determine the performance of an INIS-DDSM are as follows:

- l : the order of the modulator.
- h : the initial order of high-pass shaping of the quantization error at low frequencies.
- w : the spread of the DDSM output.
- u : the order of polynomial nonlinearities up to which the quantization error is immune from spurs.

Figure 3 compactly illustrates the role played by each parameter within the DAC.

C. INIS Conditions

If condition (20) is satisfied, the DDSM is guaranteed to be immune from spurious tones for all possible static nonlinearities with which it might interact.

To prevent nonlinearity-induced spurs, the FIR filter cannot have a linear phase, because it must compensate for the effect of the (static) nonlinearities. This leads to the following condition:

Proposition 1: A necessary condition to compensate for any static nonlinearity is that the sequence

$$\{1 \ d_1 \ d_2 \ \dots \ d_{l-1} \ d_l\} \quad (21)$$

is neither even nor odd symmetric.

Proof: The error e introduced by the floor function is filtered with an FIR filter to obtain the quantization error e_q . Our goal is to prevent the application of a nonlinear function to e_q from generating any spurious tones. In other words, we need to counteract the effect of nonlinearity. To this end, we need a distortion of the output signal of the modulator, which can be obtained through an FIR filter with a nonlinear phase. The linear phase condition for FIR filters is that the sequence of coefficients (21) is neither even nor odd symmetric [21]. Therefore, the nonlinear phase which is necessary to distort the output signal of the modulator requires necessarily that the coefficients satisfy this condition. ■

To determine the value of u , we can resort to the theory of nonsubtractive dither [15]. An example of a nonsubtractively dithered (NSD) quantizing system is given in Fig. 4. Here, a signal $v[n]$ is added to the input $x[n]$ and the sum is fed to the quantizer which provides the output $y[n]$. As discussed before, the quantizer introduces an error $e[n]$ such that:

$$y[n] = x[n] + v[n] + e[n]. \quad (22)$$

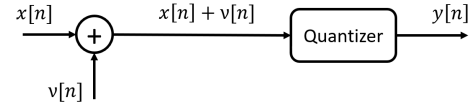


Fig. 4. Block diagram of an NSD quantizing system [15].

By comparing Eqs. (12) and (22), it is apparent that the EFM in Fig. 1 is equivalent to the NSD system in Fig. 4, provided that the following condition holds:

$$v[n] = \sum_{i=1}^l d_i e[n - i]. \quad (23)$$

At this point, we recall Theorem 6 of [15], which states that¹:

Theorem 1: Let a nonsubtractive dither signal $v[n]$ be generated by the summation of m random processes which are (i) statistically independent and (ii) uniformly distributed, with (iii) peak-to-peak amplitudes equal to integer multiples of the quantizer's least significant bit (LSB). This dither signal renders $E\{e_q^p[n]\}$ independent of the system input distribution for $p = 0, 1, \dots, m$, where $E\{\cdot\}$ denotes the expectation operator.

Owing to Corollary 3 in [15], if $m \geq w$, the last addend of Eq. (19) can be recast as

$$\sum_{p_1=1}^w \sum_{p_2=1}^w \beta_{p_1} \beta_{p_2} E[e_q^{p_1}[n]] E[e_q^{p_2}[n - k]], \quad (24)$$

meaning that $e_q^{p_1}[n]$ and $e_q^{p_2}[n]$ are uncorrelated.

Theorem 1 and Eq. (24) imply that the quantization error generated by the system is immune from spurs in the case of polynomial distortion up to order $u = m$ [22]. Owing to Eq. (23), the order m corresponds to the number of non-zero integer coefficients d_i . However, this is true only if Theorem 1 is applicable, i.e., if the error $e[n]$ (see Eq. (23)) is a statistically-independent and uniformly-distributed random process with peak-to-peak amplitude equal to integer multiples of the quantizer's LSB. Condition (iii) is satisfied because $e[n]$ ranges in the interval $(-1, 0]$ (see Eq. (3)). Nevertheless, the other two conditions are not guaranteed as is, but require the presence of a further additive 1-bit dithering, which is a common practice to prevent the generation of idle tones at the cost of a negligibly increased noise floor [23]. This technique

¹It is worth remarking that we have used a different notation compared to [15]. The total quantization error $e_q[n]$ is denoted in that work as ϵ .

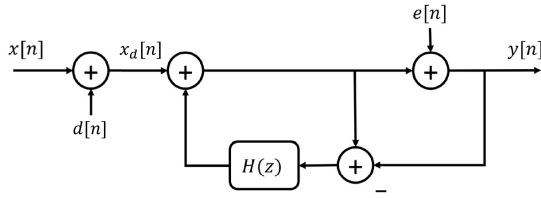


Fig. 5. Block diagram of the EFM with 1-bit dithering.

consists of adding a 1-bit random sequence to the LSB of the input. With this addition, Fig. 1 changes to Fig. 5, where $d[n]$ is the 1-bit dither signal. Experimental evidence indicates that incorporating 1-bit dithering makes the quantization error white [19], [24], [25]. The validity of conditions (i) and (ii) can be assessed by referring to Theorem 1 in [24], restated below in a more compact form:²

Theorem 2: Suppose the input to a non-overloading single-quantizer DDSM is $x_d[n] = x[n] + d[n]$, where $x[n] \in [0, 1)$ is an arbitrary bounded sequence represented with M bits and $d[n]$ is a 1-bit dither sequence. The sequence $e[n]$ is asymptotically independent, identically, and uniformly distributed and independent of $x[n]$ if and only if at least one of the following conditions is true for each integer $p > 0$, and each integer pair $(k_1, k_2) \neq (0, 0)$ and $0 \leq k_1, k_2 < N = 2^M$.

- (A) The sequence $\langle k_1 f[r] + k_2 f[r + p] \rangle_N$ does not converge to zero as $r \rightarrow \infty$, where $\langle \cdot \rangle_N$ denotes the modulo N operator.
- (B) A non-negative integer $s = s(p) \neq p$ exists such that $\langle k_1 f[s] + k_2 f[s + p] \rangle_N = N/2$.
- (C) A non-negative integer $t = t(p) < p$ exists such that $\langle k_2 f[t] \rangle_N = N/2$.

Remark 1: Theorem 2 guarantees pairwise independence, whereas, Theorem 1 requires mutual independence. In general, pairwise independence does not imply mutual independence [26], [27]. However, it has been shown that these two notions of independence are asymptotically equivalent (up to a null set) when considering a large collection of real-valued random variables [28], [29], [30], as in our case. Under an other perspective, we can interpret the pairwise independence as a first order approximation of mutual independence, also taking into account that all existing tests for high dimensional mutual independence are based on the pairwise approach [31].

In our case, the sequence $f[n]$ is

$$f[n] = \mathcal{Z}^{-1}\{F(z)\}, \quad (25)$$

where (see Appendix A)

$$F(z) = \frac{1}{1 - H(z)} = \frac{1}{1 + \sum_{i=1}^l d_i z^{-i}}. \quad (26)$$

The following (new) theorem states how u can be determined in terms of coefficients d_i .

Theorem 3: Let the coefficients d_i , with $i = 1, \dots, l$, of a DDSM be such that the sequence $f[n]$ satisfies the conditions of Theorem 2. The spur immunity order u ($= m$) of the modulator is equal to the number of coefficients d_i that are

non-zero:

$$u = \|[d_1, \dots, d_l]\|_0, \quad (27)$$

where $\|\cdot\|_0$ denotes the L_0 norm, i.e., the number of non-zero entries of the argument vector.

Proof: If the assumptions of Theorem 2 are satisfied, the error $e[n]$ is a statistically independent, identical, and uniformly distributed random variable. Moreover, $e[n]$ ranges in the interval $(-1, 0]$ (see Eq. (3)). Given Eqs. (12), (22), and (23), the DDSM is an NSD system where the nonsubtractive dither signal is given by the summation of m delayed versions of $e[n]$, being m the number of coefficients d_i that are non-zero.

Consequently, we can apply Theorem 1 to the DDSM, and obtain that $e_q^p[n]$, for $p = 1, \dots, m$ is free from spurious tones. Therefore, owing to Definition 1, for a constant input $x[n]$ the order of spur immunity u of the modulator is equal to the number m of non-zero coefficients d_i . ■

Corollary 1: The coefficients d_i of an INIS-DDSM must be 1, 0, or -1 .

Proof: To have spur immunity, the coefficients d_i must be chosen such that w (see Eq. (15)) is less than or equal to u (see Eq. (20)). Given Eq. (15) and Theorem 3, the condition (20) can be satisfied only for $u = w$, namely when

$$\|[d_1, \dots, d_l]\|_0 = \sum_{i=1}^l |d_i|. \quad (28)$$

Therefore, from Eq. (28), it can be deduced that the coefficients d_i of an INIS-DDSM must necessarily be 1, 0, or -1 . ■

In summary, to have an INIS-DDSM, the coefficients d_i must be chosen such that their sequence is neither even nor odd symmetric (Proposition 1) and:

$$\begin{cases} d_i = \{-1, 0, 1\}, \forall i = 1, \dots, l, & (\text{Corollary 1}) \\ 1 + \sum_{i=1}^l d_i = 0, & (\text{Eq. (16)}) \\ f[n] = \mathcal{Z}^{-1} \left\{ \frac{1}{1 + \sum_{i=1}^l d_i z^{-i}} \right\} \text{ satisfies Theorem 2.} \end{cases} \quad (29)$$

In the following section, we present an example of a fourth-order INIS (denoted INIS 1) with a spread of $w = 3$ that satisfies the conditions of Theorem 2 and is therefore immune to polynomial distortion up to order $u = 3$.

III. INIS 1

In this section, we analyze the INIS 1 architecture, whose high-level implementation is shown in Fig. 6. In particular, we consider an INIS 1 with constant input driving a nonlinear DAC and characterize it through numerical simulations, to verify the effects of nonidealities. Comparisons to a non-INIS architecture and an SR are also provided. The spur immunity property of INIS 1 is also demonstrated experimentally.

The PSDs in the numerical simulations were obtained using Welch's method with Hanning windows. All PSDs are shown

²Also here, we used a different notation compared to [24].

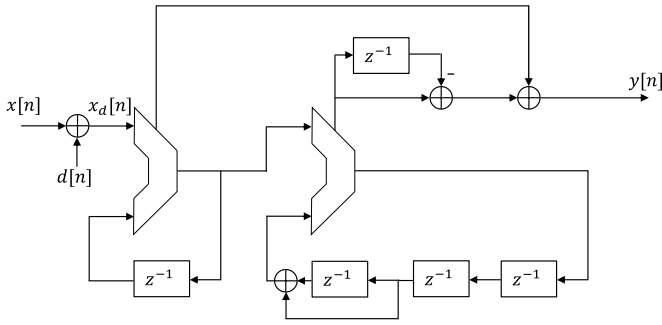


Fig. 6. Block diagram of the INIS 1 architecture.

in both dBFS and dBFS/NBW scales, i.e., decibels relative to full scale (dBFS) and dBFS referred to the noise bandwidth (NBW) specified in each figure [23].³ All the frequencies are normalized with respect to the clock frequency f_s .

A. Key Parameters

The INIS 1 modulator can be implemented with an EFM structure as in Fig. 1, with the transfer function $H(z)$ equal to:

$$H(z) = z^{-1} + z^{-2} - z^{-4}. \quad (30)$$

The order l is equal to 4, which is also the number of delay elements required to implement $H(z)$ in Eq. (30) (see Fig. 6). The modulator has the following noise transfer function:

$$NTF(z) = 1 - z^{-1} - z^{-2} + z^{-4}. \quad (31)$$

The coefficients d_i , with $i = 1, \dots, 4$, satisfy condition (16) and Corollary 1. The necessary condition of Proposition 1 is also satisfied; therefore the FIR filter expressed by $NTF(z)$ has a nonlinear phase.

Note that $NTF(z)$ has one zero at $z = 1$, which means that the order of high-pass shaping is $h = 1$. Moreover, according to Eqs. (14) and (15), the output $y[n]$ has a spread $w = 3$ and a range $[-1, 2]$. Therefore, each memoryless nonlinearity applied to $y[n]$ can be expressed as a third-order polynomial function (see Eq. (17)).

According to the analysis carried out in Section II-C, if the assumptions of Theorem 2 hold, the modulator provides an order of spur immunity $w = u = \|[-1, -1, 0, 1]\|_0 = 3$ (see Theorem 3). Consequently, according to Eq. (20), the modulator is expected to exhibit spur-free operation for any static nonlinearity $\mathcal{N}(\cdot)$ (see Sec. II-C).

B. Numerical Validation

For INIS 1, we have

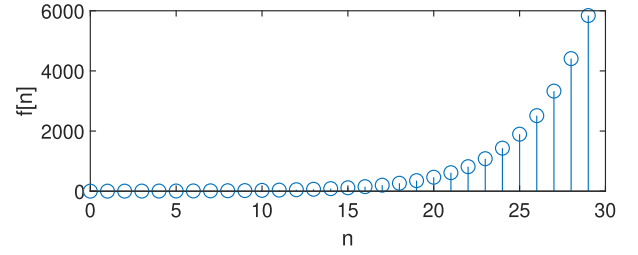
$$F(z) = \frac{1}{1 - z^{-1} - z^{-2} + z^{-4}}. \quad (32)$$

The sequence $f[n] = \mathcal{Z}^{-1}\{F(z)\}$ is shown in Fig. 7. Although it is easy to obtain a recursive expression for $f[n]$,

$$f[n] = \delta[n] + f[n-1] + f[n-2] - f[n-4], \quad (33)$$

where $\delta[n]$ is the discrete unit sample function, it is quite hard to find a non-recursive form.

³The value of NBW is calculated based on the type of windowing and the number of samples, as indicated in [23, Table A.1].

Fig. 7. Sequence $f[n]$ for the INIS 1 architecture.

For this reason, the spur immunity property is verified here by checking the validity of Theorem 2 through extensive numerical analysis up to $M = 13$. For all these cases, we verified that the assumptions of Theorem 2 are true. An analytical proof for any M is left as a challenge for the scientific community.

We carried out simulations with M ranging from 1 to 13, corresponding to N ranging from 2 to 2^{13} . We start by checking condition (C) for all strictly positive values of k_2 in each scenario. For each k_2 , we find the minimum t_{min} such that condition (C) holds. For all values of $p < t_{min}$, we check condition (A). We did not find any case where condition (A) does not hold. For $k_2 = 0$, we check condition (B) and find two values s_1 and s_2 such that it is fulfilled. We found them in all the considered cases. The simulation results are summarized in the supplementary material.

To corroborate the above results, we conducted additional tests, which further witness the robustness of our analysis.

1) *Test 1:* Following [24], as a first test, we examine only the DDSM through a simple example where the input $x[n]$ is represented using $M = 6$ bits (for visualization purposes), so that it can assume $2^6 = 64$ different values. Likewise, the error $e[n]$ is also represented with 6 bits. In our test, we use a constant input value of $1/2^6$ (for all other input values, we obtained similar results, not presented here). Our goal is to verify that the sequence $e[n]$ is independent and has a uniform distribution.

A first simulation is without dithering, and a second one with 1-bit dithering. The joint probability mass function (PMF) of $e[n]$ and its delayed version $e[n-1]$, here denoted as $P(e[n], e[n-1])$, is illustrated in Figs. 8(a) and 8(b), corresponding to the two simulations. It is evident that when 1-bit dithering is applied, the joint PMF corresponds to that of a pair of independent random variables.

Further evidence of the independence of the sequence $e[n]$ is given by its normalized autocorrelation, shown in Fig. 8(c); it indicates independence because it is significantly different from 0 only at lag 0. Finally, Fig. 8(d) shows the distribution of $e[n]$, defined as $P(e[n])$, which appears to be almost uniform.

2) *Test 2:* Test 1 is focused on Theorem 2 and provides limited information, because we fixed $M = 6$ and $x[n] = 1/2^6$. Next, we want to check the spur immunity property in the whole signal path shown in Fig. 3 through extensive simulations by changing M and $x[n]$.

First, we test the case where the modulator output is distorted by a third-order polynomial function:

$$\mathcal{N}(y) = -0.0938 + 0.945y + 0.0704y^2 - 3.33 \cdot 10^{-5}y^3, \quad (34)$$

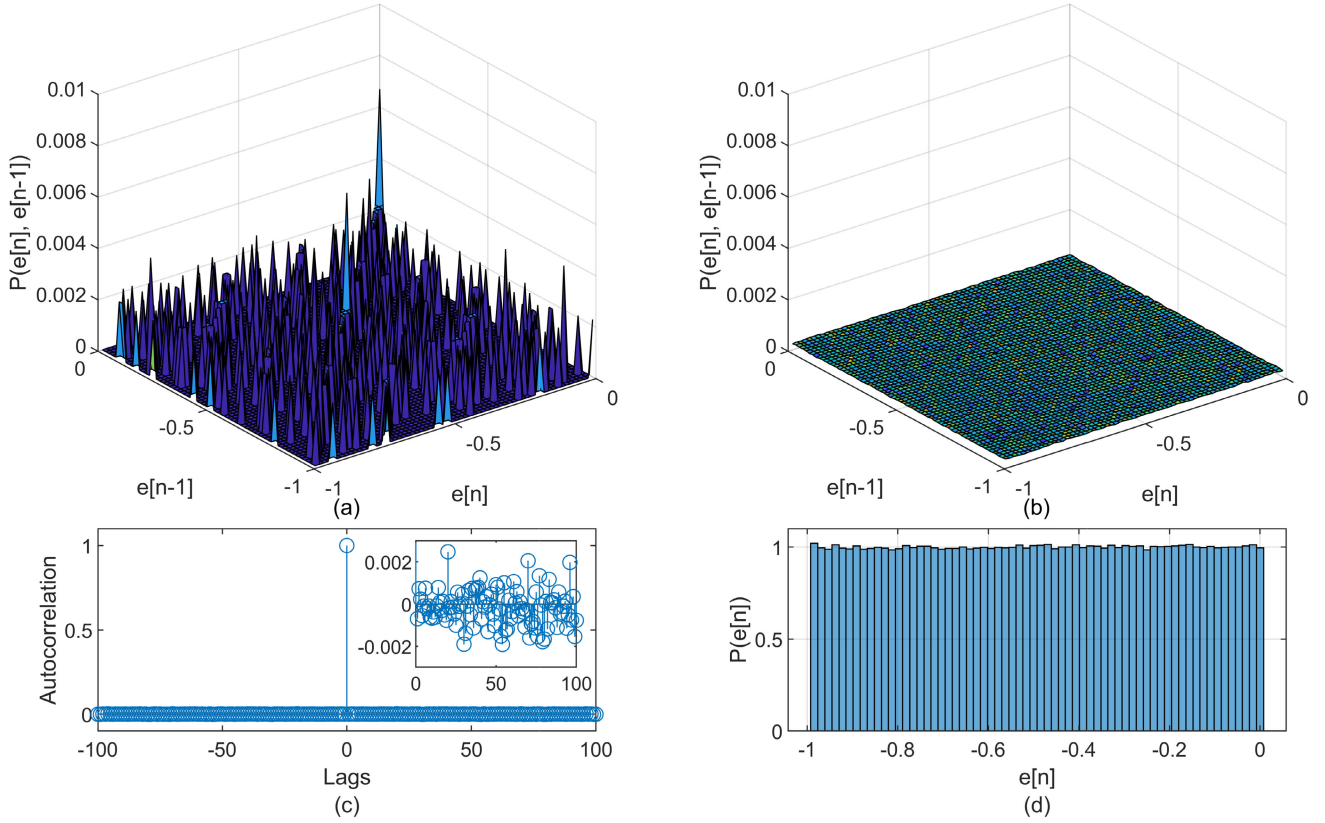


Fig. 8. Simulated joint PMF of $e[n]$ and $e[n-1]$ for INIS 1 (a) without and (b) with 1-bit dithering. (c) normalized autocorrelation and (d) PMF of $e[n]$ with dither applied.

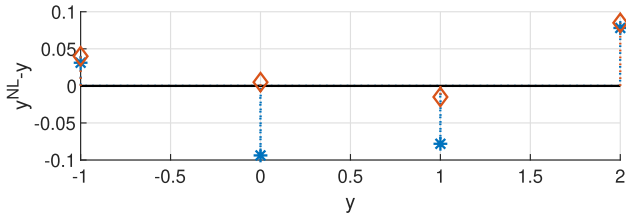


Fig. 9. INL arising from the third order polynomial (34) (blue asterisks) and the fifth order polynomial (35) (orange diamonds).

which is consistent with the integral nonlinearity (INL) shown in Fig. 9 (blue asterisks), limited to the range $[-1, 2]$. Here, the INL is defined as the point-by-point difference between y^{NL} and y [32]. The DDSM is also tested when its output interacts with a different nonlinear function, namely the fifth-order polynomial defined by

$$\mathcal{N}_1(y) = 0.005 + 0.92y + 0.01y^2 + 0.06y^3 - 0.0025y^4 - 0.0075y^5. \quad (35)$$

The INL corresponding to this polynomial is shown in Fig. 9 (orange diamonds).

The input signal $x[n]$ is coded with 12, 16, and 20 bits. In the first two cases, the system is simulated for every possible DC input, namely for 2^{12} and 2^{16} values. With 20 bits, 2^{16} DC inputs are randomly selected from the 2^{20} possible ones. In all instances, we conducted the cumulative periodogram check to detect the presence of spurious tones (see Appendix B), and the test confirmed the absence of spurs in each case.

Even though the spur immunity order u is 3 and the output is distorted by the fifth-order polynomial (35), it should not be surprising that the modulator does not exhibit spurious tones. Function (35) is indeed a fifth-order polynomial, but is evaluated only at four points, namely $\{-1, 0, 1, 2\}$. Therefore, it is possible to find a third-order polynomial that passes through the same four points. This holds true for a fifth-order polynomial but also for any other nonlinear function, as condition (20) indicates spur-free operation with any nonlinearity.

3) *Test 3:* To further test the spur immunity property of INIS 1, we randomly choose 100 input values uniformly distributed over the range $[0, 1)$ and coded with 20 bits. Moreover, 100 static INLs are chosen by randomly sampling a signal with zero mean and uniform distribution in the range $[-0.1, 0.1]$. This choice ensures that the INL stays within 10% of the LSB. For all possible combinations of input and INL, we performed the cumulative periodogram check, which indicated the absence of spurious tones.

C. Benchmarks: MASH 1-1 and SR

For comparison purposes, we also consider the case of a MASH 1-1 driving a DAC with the third-order nonlinear function (34). The NTF for the MASH 1-1 is

$$NTF(z) = (1 - z^{-1})^2 = 1 - 2z^{-1} + z^{-2}, \quad (36)$$

which has the form of Eq. (9) but does not satisfy the necessary condition of Proposition 1; therefore, it does not have the spur immunity property. Consequently, we expect that the MASH

1-1 driving a DAC with the nonlinearity described by Eq. (34) will exhibit spurs, while the INIS 1 does not.

The same DAC nonlinear function is also applied to an SR. In particular, we choose the SR described in [13, Figure 4(b)] with 20 quantization blocks, whose output is in the range $[-2, 2]$.

In order to quantify the total spur and noise power, we consider the following parameter σ (sometimes called jitter [33]):

$$\sigma = \sqrt{2 \cdot 10^{A/10}}, \quad (37)$$

where A is the integrated PSD over a normalized frequency range of interest ($[f_1, f_2]$), i.e., the area subtended by the PSD curve. If the PSD is expressed in dBFS/NBW, A can be calculated as

$$A = 10 \log_{10} \left(\int_{f_1}^{f_2} 10^{\text{PSD}/10} df \right). \quad (38)$$

The unipolar input signal $x[n]$ is coded with 20 bits.

1) *Linear DAC*: First, a linear DAC is considered. We simulated this ideal DAC, to provide a baseline comparison between INIS 1, MASH 1-1, and SR.

Figure 10(a) shows the simulated and measured PSDs of $y^L(t)$ for a constant input $x[n] = 1710/2^{20}$.

The simulated spectra of INIS 1 (blue) and SR (green) have 20 dB/decade slopes at low frequencies, while the spectrum of MASH 1-1 has a 40 dB/decade slope: MASH 1-1 has the lowest noise power at low frequencies, as expected, since it implements a higher-order ($h = 2$) noise-shaping filter. As expected, no spurious tones are present and the quantization noise introduced by the DDSM is high-pass shaped, with first-order filtering in the case of the INIS 1 and the SR and second-order filtering for the MASH 1-1. When comparing INIS 1 and SR, despite having the same filter order, it is evident that the noise level is lower for INIS.

2) *Nonlinear DAC*: Next we consider a nonlinear DAC, where the mismatch for each level is determined by the polynomial (34), corresponding to Fig. 9 (blue asterisks).

The modulator can take input values within the range $[0, 1)$. With a constant input $x[n] = X$, a MASH clocked at frequency f_s with a nonlinear DAC exhibits spurious tones at frequencies

$$f_{X1} = X f_s \quad (39)$$

and their multiples when X is small [17]. Therefore, the input ranges for which we have spurious tones within the signal bandwidth are $\left[0, \frac{1}{2 \cdot \text{OSR}}\right]$ and $\left[1 - \frac{1}{2 \cdot \text{OSR}}, 1\right)$, where OSR is the oversampling ratio. We consider $\text{OSR} = 50$, thus the ranges for which inband spurs are present are $[0, 0.01]$ and $[0.99, 1)$.

Figure 10(b) shows simulated PSDs of $y^{NL}(t)$ for the INIS 1, MASH 1-1, and SR.

As expected, with the MASH 1-1, the interaction of the DAC with the nonlinearity results in the presence of spurious tones (orange). This problem is mitigated in both the INIS 1 (blue) and SR (green); indeed, for these two architectures, no spurs are observed.

Table I shows the performance in simulation of the three modulators in response to $x[n] = 1710/2^{20}$ (within the interval

TABLE I
PERFORMANCE OF THE MODULATORS WITH CONSTANT
INPUT $x[n] = 1710/2^{20}$

Architecture	Spurs	σ [rad]
MASH 1-1	Yes	0.0203
SR	No	0.0085
INIS 1	No	0.0049

$\left[0, \frac{1}{2 \cdot \text{OSR}}\right]$, in which MASH 1-1 exhibits spurious tones). The normalized frequency range within the band $[10^{-4}, 10^{-2}]$ is considered. As expected, INIS 1 and SR do not generate spurious tones, while MASH 1-1 does. It is apparent that INIS 1 exhibits the lowest σ , corresponding to the lowest noise power.

IV. EXPERIMENTAL VALIDATION

To validate the properties of INIS observed through numerical simulations, we replicated using hardware some of the experiments carried out in simulation. This allows us to demonstrate that, even in a physical device with a nonlinear DAC, INIS does not produce spurious tones. We implemented a representative DDSM of the INIS family, specifically INIS 1, on an FPGA and compared it with a non-INIS modulator, MASH 1-1. Although experimental results are only presented for selected values of M , $x[n]$, and $\mathcal{N}(\cdot)$, spur-free behavior is observed for all inputs, resolutions, and nonlinearities that were tested.

A. Physical Implementation

The INIS 1 and MASH 1-1 DDSMs were implemented using a Digilent Eclipse Z7 FPGA board [34] with a Zmod DAC 1411 module [35]. The Eclipse Z7 contains an AMD/Xilinx Zynq-7000 XC7Z020-1CLG484C FPGA [36]. The DAC 1411 module contains an Analog Devices AD7917 14-bit DAC [37]. The 14-bit DAC is used to emulate a four-level (2-bit) DAC with prescribed nonlinearities. The nonlinearity is introduced artificially, as explained in the following. The modulator output is three bits wide, is coded as two's complement, and ranges over the interval $[-1, 2]$. This 3-bit signal is encoded into a 14-bit signal on the FPGA and then fed to the DAC. The 3-to-14-bit encoding makes the DAC almost linear at the 3-bit level and, at the same time, a variation in the output level of the modulator results in a significant change in the DAC output voltage. The four levels of the modulator are obtained by varying only two out of the fourteen bits of the DAC. The remaining bits can be set to 0 for a linear DAC, or they can be used to encode the excess nonlinearity.

The input is coded with 20 bits and the clock frequency is set to 10 MHz. The implementation of the MASH 1-1 on Zynq-7000 employs 55 lookup tables (LUTs) for logic and 67 flip-flops (FFs). For INIS 1, instead, we have 57 LUTs and 90 FFs. As expected (due to the higher value of l), the resource utilization for INIS 1 is higher than for MASH 1-1.

A 16-bit linear-feedback shift register is used to generate the 1-bit dither signal that is applied to the input of INIS 1 and MASH 1-1 [38].

The experimental PSDs were measured with a Tektronix RSA5103B real-time spectrum analyzer, in turn windowed

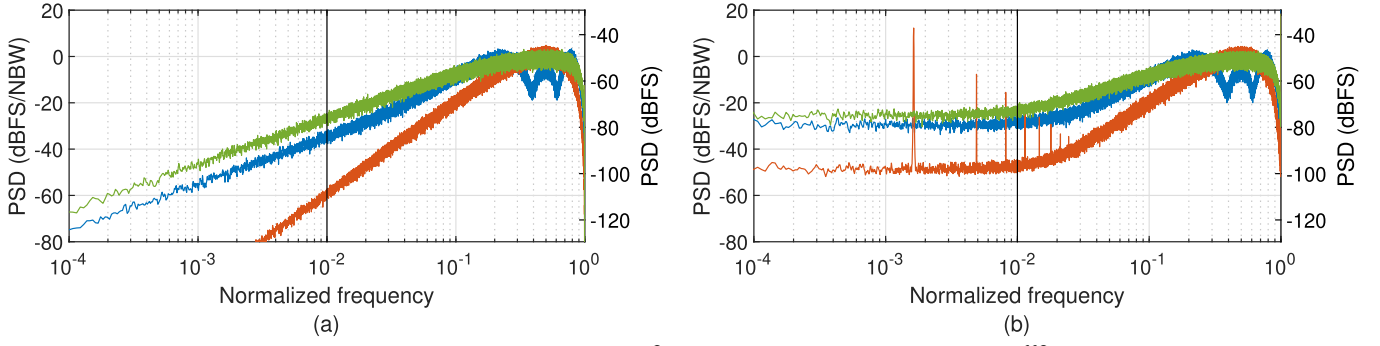


Fig. 10. Comparison of simulated PSDs of (a) the linear DAC output $y^L(t)$ and (b) the nonlinear DAC output $y^{NL}(t)$ in response to a constant input $x[n] = 1710/2^{20}$. INIS 1 (blue), MASH 1-1 (orange), and SR (green). The black vertical line marks the upper edge of the modulator bandwidth. NBW = 1.14×10^{-5} .

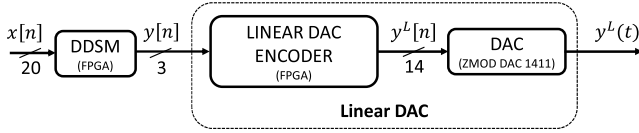


Fig. 11. Block diagram comprising a DDSM and a linear DAC.

TABLE II
LINEAR DAC ENCODING

y (binary)	y^L (binary)
-1 (111)	-1 (0000000000000000)
0 (000)	0 (0000001000000000)
1 (001)	1 (0000010000000000)
2 (010)	2 (0000011000000000)

TABLE III
NONLINEAR DAC ENCODING

y (binary)	y^{NL} (binary)
-1 (111)	-0.969 (00000000000100)
0 (000)	-0.094 (00000001110100)
1 (001)	0.922 (00000011110110)
2 (010)	2.078 (00000110001010)

Figure 12(b) shows experimental PSDs of $y^{NL}(t)$ for the INIS 1 and MASH 1-1 DDSMs in the case of a constant input, $x[n] = 1710/2^{20}$. Simulated (see Fig. 10(b)) and experimental results match well. When implemented in hardware, INIS 1 does not produce spurious tones, as expected.

with Hanning and then averaged [39]. The input impedance of the spectrum analyzer (50 Ω) matches the output impedance of the DAC.

B. Linear DAC

First, we reproduced a linear DAC in hardware. Fig. 11 shows the block diagram, where the encoder works as detailed in Tab. II.

We applied the same constant input used in Fig. 10(a) to the system, namely $x[n] = 1710/2^{20}$. Figure 12(a) shows the measured PSDs of $y^L(t)$.

The spectra flatten out at around -40 dBFS/NBW corresponding to the measurement noise floor. The experimental spectra also manifest an anomalous disturbance just below 17 kHz; this was traced to the Digilent FPGA motherboard.⁴

Apart from this anomaly, no spurious tones are present. The experimental results agree with the simulated ones (see Fig. 10(a)), the only differences being the board-related anomaly at 17 kHz and that the experimental PSDs exhibit a measurement noise floor.

C. Nonlinear DAC

Next, we consider the nonlinear DAC, whose block diagram is shown in Fig. 13. The encoder operates according to Tab. III and the mismatch for each level is determined by the polynomial (34), corresponding to Fig. 9 (blue asterisks).

⁴The disturbance present in the measurements around 17 kHz is due to the FPGA and is independent of the modulator. Since this disturbance (very low power, less than 100 dBm) is consistently present, it does not affect the comparison between the DDSMs.

V. HIGHER-ORDER INIS-DDSMs

INIS 1 has $h = 1$, meaning that it filters quantization error using a first-order filter. However, it is also possible to implement INIS modulators with higher-order filters. This section provides an analysis of two of these, referred to as INIS 2 and INIS 3, and includes a comparison with SR.

The signals $x[n]$ that we consider are coded with 20 bits.

A. INIS 2

As a second example of the INIS family, in this section we briefly analyze the architecture INIS 2.

Also in this case, the modulator can be implemented using an EFM structure with

$$H(z) = z^{-1} - z^{-2} + z^{-3} + z^{-4} - z^{-6}. \quad (40)$$

The coefficients d_i , with $i = 1, \dots, 6$, satisfy condition (16), Proposition 1, and Corollary 1.

The order l is equal to 6, the order of high-pass shaping is $h = 2$, and the output spread is $w = 5$ over a range $[-2, 3]$. Therefore, each static nonlinearity after the DDSM can be expressed as a polynomial function of order five or higher. The order of spur immunity is $u \geq 5$, meaning that the spectrum of the quantization error is immune from spurs at least up to the term e_q^5 . This property will be validated in the case where the DDSM output is distorted by:

$$\begin{aligned} \mathcal{N}_2(y) = & -0.0128 + 0.9076y + 0.01515y^2 \\ & + 0.05153y^3 - 0.001804y^4 - 0.004824y^5. \end{aligned} \quad (41)$$

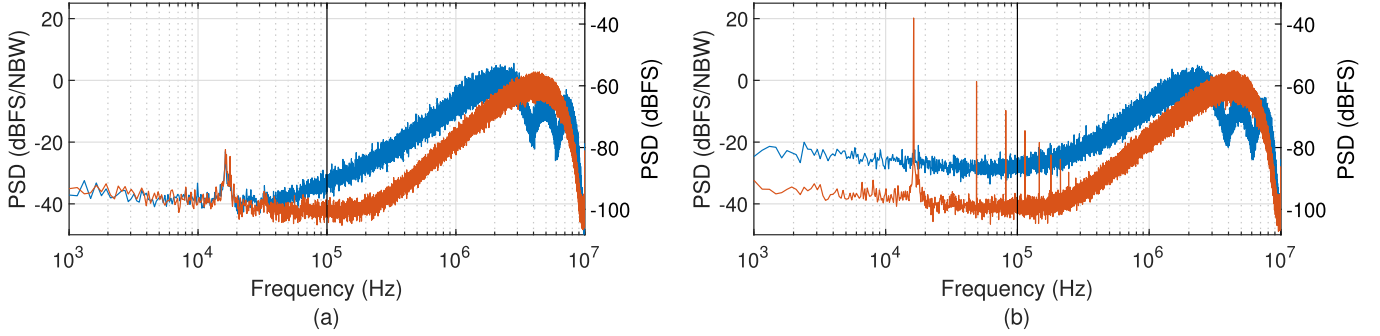


Fig. 12. Comparison of experimentally measured PSDs of (a) the linear DAC output $y^L(t)$ and (b) the nonlinear DAC output $y^{NL}(t)$ in response to a constant input $x[n] = 1710/2^{20}$. INIS 1 (blue) and MASH 1-1 (orange). The black vertical lines mark the upper edge of the modulator bandwidth. NBW = 1.50×10^{-6} .

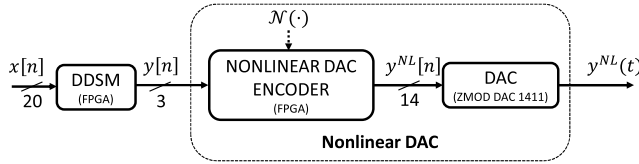


Fig. 13. Block diagram comprising a DDSM and a nonlinear DAC.

The INL corresponding to this polynomial is shown in Fig. 14(a).

Similar to INIS 1, the validity of Theorem 2 is demonstrated with extensive simulations up to $M = 12$ (see supplementary material).

B. INIS 3

We also consider the INIS 3 architecture, which is defined by

$$H(z) = z^{-1} - z^{-2} + z^{-3} + z^{-4} + z^{-5} - z^{-6} - z^{-7} - z^{-8} + z^{-9} - z^{-10} + z^{-11}. \quad (42)$$

Also in this case, condition (16), Proposition 1, and Corollary 1 are satisfied by the coefficients d_i , with $i = 1, \dots, 11$.

It has order $l = 11$, order of high-pass shaping $h = 3$, and output spread $w = 11$, within the range $[-5, 6]$. Each memoryless nonlinearity applied to the INIS 3 output can be expressed as a eleventh-order polynomial function, whose INL is shown in Fig. 14(b). Also with this DDSM, the validity of Theorem 2 is demonstrated through simulations up to $M = 12$ (see supplementary material).

C. Benchmark: SR

The performance of the INIS-DDSMs is next compared to the SR introduced in Section III-C.

SRs of the first order are proven to be spur immune from nonlinearity in both the quantization error and the accumulated quantization error [13]. However, SRs of higher order are spur-immune only in the accumulated quantization error, which makes them suitable for frequency synthesizers but not for DACs [40], [41]. This is the reason why an SR with $h = 1$ is used as a benchmark.

For a fair comparison, the same nonlinear function $\mathcal{N}_2(\cdot)$ applied to INIS 2 is also applied to the SR.

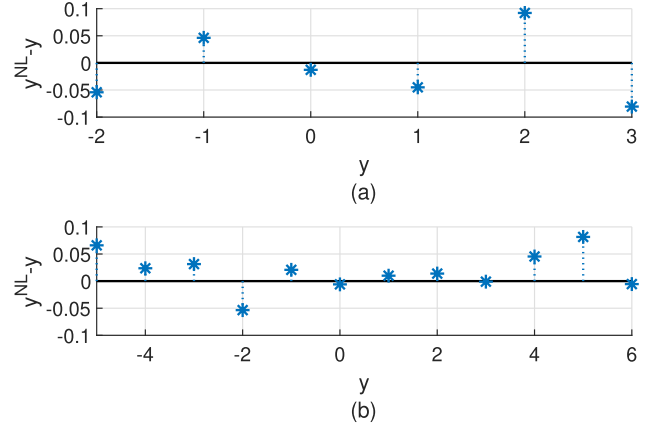


Fig. 14. INL arising from (a) the fifth order polynomial (41) used with INIS 2 and (b) the eleventh order polynomial used with INIS 3.

TABLE IV
COMPARATIVE PERFORMANCE OF SR, INIS2 AND INIS 3 MODULATORS WITH CONSTANT INPUT $x[n] = 1710/2^{20}$

Architecture	σ [rad]
SR	0.0048
INIS 2	0.0031
INIS 3	0.0024

We consider the same input used in Sec. III-C, that is $x[n] = 1710/2^{20}$. In Fig. 15, the comparison between the distorted spectra of INIS 2 (blue curves), INIS 3 (orange curves), and SR (green curves) is shown. Panel (a) shows the PSD of y^{NL} for INIS 2 (blue), INIS 3 (orange), and SR (green).

The quantization error e_q is independent of the nonlinearity introduced by the DAC, and results from a high-pass shaping filtering of order h . This means that its spectrum has a slope equal to $20h$ dB/dec (see Fig. 3). The higher is h , the lower is the power of e_q within the modulator's band. Figure 15(b) shows the PSD of e_q (same color code as in panel (a)), evidencing that $h = 1$ for SR, $h = 2$ for INIS 2, and $h = 3$ for INIS 3.

The performance of the modulators, expressed in terms of σ , are shown in Table IV.

None of the modulators tested exhibit spurs, as expected. INIS 3 performed the best in terms of noise power, followed by INIS 2 and SR. It is clear that the noise power depends on the modulator's input and the specific nonlinear function applied to the modulator's output. However, while SR has the property of spur immunity for the quantization error with $h = 1$ by

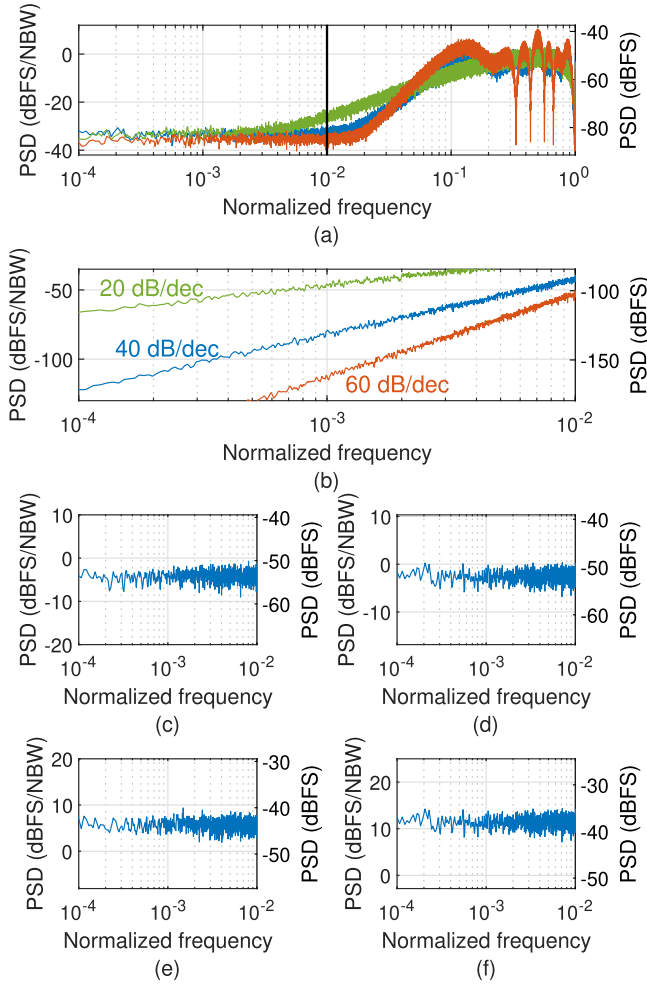


Fig. 15. Spectral features of the DDSM response to $x[n] = 1710/2^{20}$. PSD of (a) y^{NL} and (b) e_q : INIS 2 (blue curves), SR (green curves), and INIS 3 (orange curves). PSD of (c) e_q^2 , (d) e_q^3 , (e) e_q^4 , and (f) e_q^5 for INIS 2. NBW = 1.14×10^{-5} .

definition [13], it is possible to have INIS-DDSMs with the same property of spur immunity, but with $h > 1$. This means that it is possible to have INIS-DDSMs where the noise power of the quantization error e_q is consistently lower than that with SR, regardless of the nonlinearity.

VI. CONCLUSION

In this paper, we have introduced a family of $\Delta\Sigma$ modulators that inherently achieve spur-free operation, independently of the static nonlinearity encountered. The architecture and design constraints are described. An example of an INIS architecture is presented and compared with a traditional MASH-DDSM and with an SR. We have shown that the spectrum of an INIS-DDSM with polynomial distortion exhibits a folded noise floor similar to that of a MASH-DDSM but without spurs, unlike the MASH-DDSM. In addition, while not exhibiting spurious tones like SR, the INIS-DDSM exhibits lower in band noise power compared to SR.

The immunity of three examples of the INIS family to spurs has been proven with a brute force approach when the DDSM buses are coded up to a certain number of bits. The spur immunity property is also validated through extensive numerical analyses and with an experimental demonstrator.

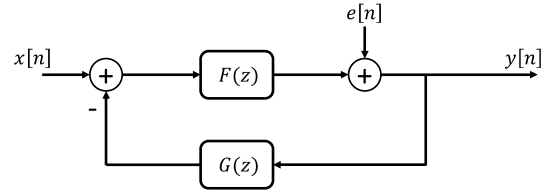


Fig. 16. Block diagram of the DDSM used in [24].

There is currently no general analytical method to prove the immunity to spurious tones. This remains an open problem and can be an interesting challenge for the scientific community.

APPENDIX A

Figure 16 shows the DDSM diagram used in [24]. With this scheme, the Z-transform of the output $y[n]$ is given by [24]:

$$Y(z) = \frac{F(z)}{1 + F(z)G(z)}X(z) + \frac{1}{1 + F(z)G(z)}\mathcal{E}(z) \quad (43)$$

To make the diagrams in Figs. 1 and 16 equivalent, we need to make Eqs. (4) and (43) equal. By doing so and also considering that $NTF(z) = 1 - H(z)$ (see Eq. (8)), we obtain:

$$\begin{cases} \frac{F(z)}{1 + F(z)G(z)} = 1, \\ \frac{1}{1 + F(z)G(z)} = 1 - H(z). \end{cases} \quad (44)$$

By solving the system (44), we obtain:

$$\begin{cases} F(z) = \frac{1}{1 - H(z)}, \\ G(z) = H(z). \end{cases} \quad (45)$$

APPENDIX B

The cumulative periodogram check is a statistical test used to evaluate whether a sequence exhibits periodicity [42]. The periodogram estimates the PSD of the sequence, and the cumulative periodogram is the sum of the normalized PSD values across a range of frequencies of interest.

Let us consider a constant input $x[n]$ to the modulator. The output of the DDSM distorted by the nonlinearity, $y^{NL}[n]$, exhibits periodicity only if spurious tones are present. In this scenario, the cumulative periodogram check can be applied to $y^{NL}[n]$ to detect the presence of spurious tones.

In the absence of spurious tones, the PSD of $y^{NL}[n]$ within the DDSM bandwidth is expected to resemble that of white noise. As a result, the cumulative periodogram should approximate a straight line. This straight line is taken as a reference. Next, to account for statistical variations, a confidence boundary above the reference is established. For this analysis, a 95% confidence level is used, which is a standard value for such statistical tests [42].

The cumulative periodogram of $y^{NL}[n]$ is compared to the confidence boundary. If the cumulative periodogram exceeds the boundary at any frequency, the test indicates the presence of spurious tones in the DDSM output.

As an example, Figure 17(a) shows the PSD of $y^{NL}[n]$ when $x[n] = 2/2^{12}$ is used as an input to INIS 1. The bandwidth of the DDSM is shown with the black vertical line. Figure 17(b) shows the cumulative periodogram

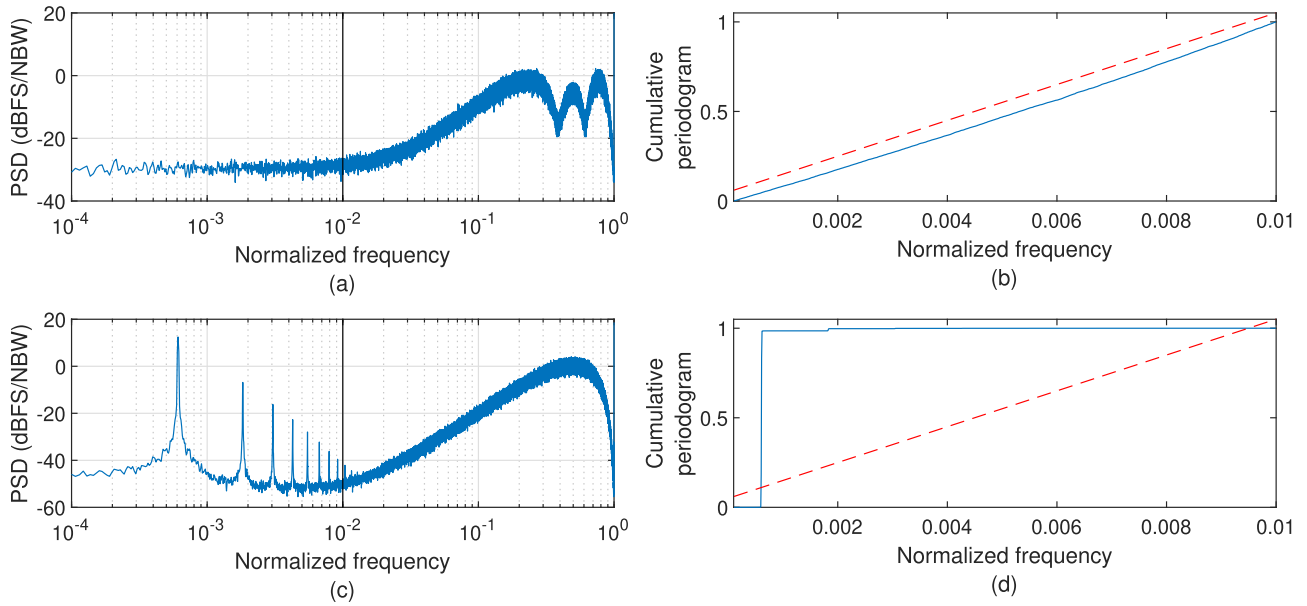


Fig. 17. INIS 1: (a) PSD of y^{NL} (blue line); (b) cumulative periodogram (blue line). MASH 1-1: (c) PSD of y^{NL} (blue line); (d) cumulative periodogram (blue line).

computed within the DDSM bandwidth (blue line) and the 95%-confidence boundary (red line). INIS 1 is spur free, and the cumulative periodogram is below the confidence boundary. The same scenario is replicated with MASH 1-1 in Figs. 17(c) and 17(d). MASH 1-1 exhibits spurious tones, and the cumulative periodogram exceeds the confidence boundary.

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Alessandro Ravera (Member, IEEE) received the B.Sc. degree in electronic engineering and information technology and the M.Sc. degree in electronic engineering from the University of Genoa, Italy, in 2019 and 2021, respectively, where he is currently pursuing the Ph.D. degree. His current research interests are in the modeling of nonlinear systems and the design and implementation of digital embedded systems.



Valerio Mazzaro was born in Salerno, Italy, in 1991. He received the B.Sc. and M.Sc. degrees in electronic engineering from the Università di Napoli Federico II, Naples, Italy, in 2014 and 2017, respectively, and the Ph.D. degree from the University College Dublin, Dublin, Ireland, in 2023. In 2022, he joined STMicroelectronics, Pavia, Italy. His research interests include frequency synthesizers, digital delta-sigma modulators, and phase-locked loops.



Marco Storace (Senior Member, IEEE) was born in Genoa, Italy, in 1969. He received the Laurea (M.Sc.) degree (summa cum laude) in electronic engineering and the Ph.D. degree in electrical engineering from the University of Genoa, in March 1994 and April 1998, respectively. Since 2011, he has been a Full Professor with the University of Genoa. He was a Visitor with EPFL, Lausanne, Switzerland, in 1998 and 2002. He has authored or co-authored about 180 scientific articles, more than an half of which have been published in international journals. He has co-authored *Linear and Nonlinear Circuits: Basic & Advanced Concepts* (2 volumes). His current research interests are in the area of nonlinear circuit theory and applications, analysis, and modeling of nonlinear dynamical systems, also networked. He is a member of the IEEE Technical Committee on Nonlinear Circuits and Systems (TC-NCAS). He served as an Associate Editor for IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS—II: EXPRESS BRIEFS (2008–2009) and as the Chair for the Nonlinear Circuits and Systems Technical Committee (2021–2023).



Michael Peter Kennedy (Fellow, IEEE) received the B.E. degree in electronics from the National University of Ireland, Dublin, in 1984, and the M.S. and Ph.D. degrees from the University of California at Berkeley (UC Berkeley), Berkeley, in 1987 and 1991, respectively, the D.Eng. degree from the National University of Ireland in 2010, and the Honorary D.Sc. degree in engineering from the Queen's University of Belfast in 2020. He was a Design Engineer with Philips Electronics, a Post-Doctoral Research Engineer with the Electronics Research Laboratory, UC Berkeley, and as a Professeur Invité with the Federal Institute of Technology Lausanne (EPFL), Switzerland. From 1992 to 2000, he was a Faculty Member of the Department of Electronic and Electrical Engineering, University College Dublin (UCD), Dublin, Ireland, where he taught electronic circuits and computer-aided circuit analysis and directed the Undergraduate Electronics Laboratory. In 2000, he joined University College Cork (UCC), Cork, Ireland, as a Professor and the Head of the Department of Microelectronic Engineering. He was the Dean of the Faculty of Engineering, UCC, from 2003 to 2005, and the Vice-President of Research from 2005 to 2010. He returned to UCD as a Professor in microelectronic engineering in 2017, where he is currently the Head of Electronic Engineering. He was the Founding Director of Ireland's Microelectronics Industry Design Association in 2001 and has been the Scientific Director of the Microelectronic Circuits Centre Ireland since 2010. He has over 400 publications in the area of nonlinear circuits and has taught courses on nonlinear dynamics and delta-sigma modulation in China, England, Greece, Hungary, Italy, South Korea, Spain, Switzerland, and USA. His research interests are in the simulation, analysis and design of nonlinear dynamical systems for applications in communications, and signal processing. He was made an IEEE Fellow in 1998 for his contributions to the study of Neural Networks and Nonlinear Dynamics. He was a recipient of the 1991 Best Paper Award from the *International Journal of Circuit Theory and Applications* and the Best Paper Award at European Conference on Circuit Theory and Design 1999. He served as an Associate Editor for IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS from 1993 to 1995 and from 1999 to 2004. He was awarded the IEEE Third Millennium Medal, the IEEE Circuits and Systems Society Golden Jubilee Medal in 2000, and the inaugural Parson's Medal for Engineering Sciences by the Royal Irish Academy (RIA) in 2001. He was elected to membership of the RIA in 2004, served as a RIA Policy and International Relations Secretary from 2012 to 2016, and as the President from 2017 to 2020. He was the Vice-President for Region 8 of the IEEE Circuits and Systems Society (CASS) from 2005 to 2007, a CASS Distinguished Lecturer in 2012 and 2013 and again 2022 and 2023, and the Chair of the CASS Distinguished Lecturer Program from 2017 to 2020. He served on the IEEE Fellows Committee, the IEEE Gustav Robert Kirchhoff Award Committee, and the IEEE Technical Field Awards Committee. He is currently Chair of the IEEE Awards Board.